

PneumoNet

*Academic Research & Productionization Artifact — Attention-Guided, Explainable Multi-Class
Chest X-Ray Classification*

Working scope: Normal • Pneumonia • Tuberculosis • COVID-19

Final-year Computer Engineering / Data Science research project



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1. Executive Overview

PneumoNet is proposed as a research-oriented deep-learning system for four-class chest X-ray (CXR) classification: Normal, Pneumonia, Tuberculosis (TB), and COVID-19. The project is deliberately structured as both a scientific study and a production-oriented machine-learning system.

The central research idea is not simply to build an EfficientNet model or to prove that attention is useful. Instead, the project establishes CNN baselines, introduces attention mechanisms under controlled conditions, performs ablation studies, evaluates classification and class-wise performance, investigates Grad-CAM explanations, and—where suitable data permit—tests external/cross-dataset generalization. The eventual research contribution is derived from the weakness revealed by those experiments.

This framing matters because prior literature already contains four-category CXR work involving COVID-19, Pneumonia, TB, Normal and XAI/Grad-CAM. Therefore, merely combining these components would not by itself constitute novelty. A 2022 study, for example, used 7,132 CXR images across four categories and employed Grad-CAM, LIME and SHAP. [citespace.org/turn0search2/turn0search4](#)

The production side then turns the validated research model into a reproducible ML system using experiment tracking, model versioning, an inference API, containerization, automated testing/CI-CD, and cloud deployment. The deployed system is a research/decision-support prototype, not an autonomous clinical diagnostic device.

2. Proposed Title

Primary working title: Attention-Guided and Explainable Deep Learning for Robust Multi-Class Chest X-Ray Classification

Working product/repository name: PneumoNet

Important academic note: the name PneumoNet should be treated as a working name until a final literature and publication-name check is completed. The paper title should emphasize the research question and contribution rather than the product name.

3. Introduction

3.1 Background

Chest radiography is a widely used medical imaging modality and has become a major research area for deep-learning-based image analysis. Existing literature covers image-level classification, localization, segmentation, image generation, and domain adaptation. A major survey of CXR deep learning explicitly organizes the field around these task categories and also highlights caveats surrounding public datasets and clinical usefulness. [citespace.org/turn0search0/turn0search1](#)

Convolutional neural networks (CNNs) are effective at learning hierarchical visual representations directly from images. Transfer learning allows a CNN pretrained on a large image dataset to be adapted to CXR classification, reducing the amount of data and training time required compared with training a deep model entirely from scratch.

However, high test accuracy alone is insufficient for a research-grade medical-imaging system. Dataset composition, class imbalance, label quality, acquisition differences, domain shift, leakage, calibration, and



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interpretability can materially affect conclusions. A systematic review of external validation in radiologic deep learning found that most evaluated algorithms showed reduced performance on external datasets compared with internal development data. 穀cite學turn0search7傢

3.2 Clinical/technical motivation

- Several thoracic diseases can exhibit overlapping radiographic patterns.
- A classifier can perform well overall while performing poorly on a specific disease class.
- Public datasets can differ in acquisition conditions, patient populations, labeling processes, image quality and class distributions.
- Deep-learning predictions are difficult to inspect directly because the learned representation is distributed across many parameters.
- Grad-CAM can provide a visual attribution map, but it is a coarse explanation rather than proof of lesion localization.
- A research model should be reproducible and traceable from dataset version through experiment to deployed model.

4. Problem Statement

Develop and experimentally validate a reproducible deep-learning framework capable of classifying chest X-ray images into Normal, Pneumonia, Tuberculosis, and COVID-19 while systematically investigating the effect of CNN architecture and attention mechanisms, analyzing prediction explanations with Grad-CAM, and assessing whether observed improvements generalize beyond the development distribution.

The project additionally aims to productionize the validated model as an ML service with experiment tracking, model versioning, containerization and API-based inference.

5. Research Gap

The project should not claim that attention, EfficientNet, or Grad-CAM are individually novel. Existing literature already contains multi-class CXR classification with these kinds of components. For example, the 2022 explanatory CXR study classified four categories and used Grad-CAM/LIME/SHAP.

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A stronger gap is therefore framed around controlled evaluation and robustness:

- How much does attention improve over a matched CNN baseline rather than an unrelated baseline?
- Does the benefit of attention depend on the backbone architecture?
- Which attention component actually contributes to any observed improvement?
- Do improvements persist across disease classes rather than only in aggregate accuracy?
- Do attention improvements persist on an external or cross-domain dataset?
- Do Grad-CAM explanations become more anatomically relevant, or merely look visually different?



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- Can the model and experiments be made sufficiently reproducible for production-oriented use?

These are research questions to test, not conclusions to assume.

6. Aim

To design, evaluate and productionize an explainable attention-guided deep-learning framework for robust four-class chest X-ray classification, with emphasis on controlled experimentation, ablation, explainability and external generalization.

7. Objectives

1. Review the CXR deep-learning, transfer-learning, attention, XAI and domain-generalization literature.
2. Select and document appropriate public CXR datasets and their class definitions.
3. Perform dataset quality assessment, leakage checks and class-distribution analysis.
4. Build a reproducible preprocessing and augmentation pipeline.
5. Train representative CNN baselines such as ResNet, DenseNet and EfficientNet.
6. Add channel and/or spatial attention mechanisms under controlled conditions.
7. Perform ablation studies to isolate the contribution of attention components.
8. Evaluate accuracy, precision, recall, macro-F1, AUROC and class-wise metrics.
9. Use Grad-CAM to analyze prediction attribution.
10. Where possible, evaluate internal versus external/cross-dataset generalization.
11. Use the experimental findings to identify and validate a targeted research contribution.
12. Track experiments and artifacts with MLflow and version the final model.
13. Expose the validated model through a FastAPI inference service and package it with Docker.
14. Implement automated testing and deployment practices and deploy the research prototype to a cloud environment.

8. Research Questions and Hypotheses

RQ	Question	Hypothesis
RQ1	Which CNN backbone gives the best performance/efficiency trade-off?	Performance will vary across ResNet, DenseNet and EfficientNet.
RQ2	Does attention improve matched CNN baselines?	At least one attention configuration will improve one or more important metrics.
RQ3	Does attention benefit depend on the backbone?	The effect size will vary across architectures.



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RQ4	Which attention component matters?	Ablation will reveal whether channel, spatial or combined attention contributes most.
RQ5	Do explanations change with attention?	Attention models will produce different attribution patterns from their baselines.
RQ6	Do gains generalize externally?	Internal gains may shrink under domain shift.
RQ7	Can the identified weakness be addressed?	A targeted modification based on findings will improve the selected research metric without unacceptable trade-offs.

9. Proposed End-to-End Solution

The system has two tightly connected tracks: a research track and a production track.

9.1 Research track

Dataset → quality assessment → preprocessing → CNN baselines → attention variants → ablation → evaluation → Grad-CAM → external validation → research finding → proposed modification → final validation.

9.2 Production track

Validated model → MLflow model/artifact tracking → model registry → FastAPI inference service → Docker container → automated tests/CI-CD → cloud deployment → web interface.

10. Dataset Strategy

10.1 Candidate data sources

Candidate sources may include established CXR datasets such as NIH ChestX-ray14, TBX11K, COVIDx and other suitable public collections. The final selection must be based on class definitions, licensing/usage conditions, patient metadata, label reliability and compatibility with the four-class research question rather than simply on availability.

The literature includes multiple large public CXR datasets and explicitly warns that dataset provenance and public-dataset caveats matter when interpreting results. 穀cite學turn0search0像

10.2 Dataset harmonization

- Map source-specific labels to the four project classes only when definitions are sufficiently compatible.
- Preserve the original dataset/source identifier.
- Preserve patient-level identifiers when available.
- Avoid splitting images from the same patient across train and test.
- Check for duplicate or near-duplicate images across sources.
- Record acquisition/source metadata where available.



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- Do not silently treat dataset absence of a label as evidence that a disease is absent.

10.3 Dataset leakage

A particularly dangerous failure mode is data leakage. If related images from the same patient, duplicated images, or source-specific artifacts appear in both training and test data, the reported performance may be artificially inflated.

10.4 Class imbalance

- Report class counts before training.
- Use stratified splits where appropriate.
- Consider weighted loss or balanced sampling.
- Evaluate macro-F1 and per-class recall rather than accuracy alone.
- Report sensitivity/recall for each disease class.

11. Data Preprocessing

Stage	Purpose	Example implementation
Validation	Reject corrupt/unsupported files	PIL/OpenCV checks
Resize	Standardize input dimensions	224×224 / 300×300 depending on model
Normalization	Match pretrained model expectations	Model-specific normalization
Augmentation	Improve robustness	Flip where justified, small rotations/crops, intensity variation
Balancing	Reduce class dominance	Weighted loss / balanced sampling
Metadata	Trace source and patient	CSV/Parquet metadata table

CLAHE and other enhancement techniques can be experimentally evaluated, but should not be inserted into the final pipeline simply because they are common. If preprocessing changes performance, it should be reported as an experimental factor.

12. Deep Learning Foundations Used in the Project

12.1 CNN

A convolutional neural network learns spatial feature representations using convolutional filters. Early layers tend to learn low-level patterns such as edges and textures, while deeper layers learn increasingly abstract structures.

12.2 Transfer learning

A pretrained backbone is initialized using weights learned on a large dataset and then adapted to the CXR task. This reduces training cost and can improve convergence when medical datasets are limited.

12.3 ResNet

ResNet introduces residual/skip connections that make very deep networks easier to optimize. It is useful



as a conventional strong baseline.

12.4 DenseNet

DenseNet connects each layer to later layers within a dense block, encouraging feature reuse and efficient gradient propagation. DenseNet variants have historically been widely used in CXR research.

12.5 EfficientNet

EfficientNet scales network depth, width and input resolution in a coordinated way to achieve favorable accuracy/compute trade-offs. EfficientNet is a candidate backbone, not an assumed final model.

12.6 Classification head

The pretrained feature extractor is followed by pooling and a task-specific classification layer producing four class probabilities. Softmax is appropriate for mutually exclusive four-class classification.

13. Attention Mechanisms

13.1 Why attention?

A CNN may learn features from the entire image, including irrelevant regions. Attention mechanisms attempt to reweight learned representations so that informative channels and/or spatial regions receive greater emphasis.

13.2 Channel attention

Channel attention estimates which feature channels are useful and scales them accordingly. Squeeze-and-Excitation (SE) is a representative channel-attention mechanism.

13.3 Spatial attention

Spatial attention estimates which locations in a feature map are important. It can complement channel attention by emphasizing relevant spatial regions.

13.4 Combined attention / CBAM-style design

A CBAM-style block combines channel and spatial attention sequentially. This is a candidate experimental configuration. The research question is whether the additional complexity actually improves relevant metrics.

13.5 Important academic rule

Do not say 'attention improves diagnosis' before experimentation. Say 'the study investigates whether attention improves classification performance and attribution behavior.'

14. Experimental Matrix

Experiment	Backbone	Attention	Purpose
B1	ResNet	None	Baseline
B2	DenseNet	None	Baseline
B3	EfficientNet	None	Baseline
A1	ResNet	Channel	Attention effect
A2	DenseNet	Channel	Attention effect
A3	EfficientNet	Channel	Attention effect



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A4	EfficientNet	Spatial	Attention effect
A5	EfficientNet	Channel + Spatial	Combined attention
AB1	Best backbone	Ablation	Component contribution

The final matrix should be expanded or reduced based on compute budget and literature findings. The comparison must keep the data split, preprocessing, evaluation protocol and reporting standards consistent.

15. Ablation Study

Ablation is essential because a model with multiple components cannot demonstrate which component caused an improvement. For example:

- EfficientNet
- EfficientNet + channel attention
- EfficientNet + spatial attention
- EfficientNet + channel + spatial attention

If the combined model improves only because of a preprocessing change or increased parameter count, the ablation should expose that limitation.

16. Training Strategy

- Start with frozen pretrained backbone and train the classification head.
- Gradually unfreeze deeper layers for fine-tuning.
- Use a validation set for hyperparameter selection.
- Use early stopping and learning-rate scheduling where appropriate.
- Use class-weighted loss or balanced sampling when justified by the class distribution.
- Fix random seeds where practical and report them.
- Record optimizer, learning rate, batch size, number of epochs, scheduler and augmentation settings.

16.1 Loss functions

The baseline loss for mutually exclusive four-class classification is categorical cross-entropy. If class imbalance is substantial, class-weighted cross-entropy or focal loss can be investigated as an additional controlled experiment. The choice should be justified by observed data rather than inserted by default.

16.2 Optimizers

Adam/AdamW are practical starting points for transfer learning. SGD with momentum can also be used as a comparative optimizer if computational budget allows.



17. Evaluation Metrics

Metric	Why it matters
Accuracy	Overall proportion of correct predictions; can be misleading under imbalance.
Precision	How many predicted positives for a class are correct.
Recall/Sensitivity	How many true examples of a class are detected.
F1	Balances precision and recall.
Macro-F1	Gives equal weight to each class; important for imbalance.
AUROC	Measures ranking/discrimination across thresholds; interpret carefully in multiclass settings.
Confusion matrix	Shows class-specific error patterns.
Inference latency	Relevant for production deployment.
Model size/parameters	Relevant to deployment cost.

18. Grad-CAM and Explainability

18.1 What Grad-CAM does

Grad-CAM uses gradients associated with a target prediction and spatial feature maps from a convolutional layer to produce a coarse localization/attribution heatmap. It helps visualize which image regions contributed to the prediction.

18.2 What Grad-CAM does NOT prove

- It does not prove that the highlighted region is a true lesion.
- It does not replace radiologist interpretation.
- A visually plausible heatmap does not automatically mean the model is clinically correct.
- Its spatial resolution is limited by the selected convolutional feature maps.

18.3 Proposed experiment

Generate Grad-CAM for matched baseline and attention models across true-positive, false-positive, true-negative and false-negative cases. Compare patterns across disease classes and, if feasible, quantify localization relevance using available anatomical masks or a defensible proxy metric.

19. External Validation and Domain Shift

Internal validation answers whether the model performs on data resembling its training distribution. External validation asks whether the model works on a different source/distribution. This distinction is critical in medical imaging.

A systematic review of external validation reported that 81% of evaluated radiologic deep-learning algorithms showed at least some reduction in external performance, with a substantial decrease reported in 24% of cases. 穀cite學turn0search7像



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19.1 Proposed protocol

- Train using the development dataset(s).
- Freeze the final candidate before external evaluation.
- Evaluate on an unseen source/dataset.
- Report internal and external metrics side by side.
- Analyze which classes suffer the largest degradation.
- Use Grad-CAM to inspect whether attribution patterns change across domains.

19.2 Why this can strengthen the contribution

A model that improves only on an internal test set may be exploiting dataset-specific characteristics. Demonstrating improved external robustness is a stronger claim than demonstrating a small internal accuracy increase.

20. Possible Research Contributions

The exact contribution must be selected after the baseline experiments and literature review. Candidate contribution directions include:

Observed problem	Potential contribution
Attention improves internal F1 but not external F1	Investigate and develop a domain-robust training strategy.
Attention helps only certain classes	Class-aware attention/loss strategy.
Grad-CAM focuses on irrelevant regions	Anatomically guided or artifact-aware training/evaluation.
One backbone benefits strongly from attention	Systematic backbone-attention interaction study.
Preprocessing causes major variation	Robust preprocessing/augmentation study.
Calibration is poor despite high accuracy	Calibration-aware prediction and uncertainty reporting.

The final paper should claim only the contribution that is actually demonstrated and compared against appropriate baselines.

21. Advantages

- Combines research experimentation with production engineering.
- Uses controlled baselines and ablation rather than a single arbitrary architecture.
- Addresses explainability through Grad-CAM.
- Can investigate external generalization rather than relying solely on a random internal split.
- Produces a reproducible experiment trail through MLflow.
- Demonstrates practical deployment skills through FastAPI, Docker and cloud infrastructure.
- Creates a strong academic narrative: problem → hypothesis → experiment → finding → contribution →



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validation.

22. Disadvantages / Risks

- Medical datasets can be difficult to harmonize and may have inconsistent labels.
- Combining datasets can introduce severe domain and source bias.
- High compute requirements may limit the number of experiments.
- Grad-CAM can be overinterpreted.
- A strong internal score may not translate to external performance.
- A four-class model is a simplification of real-world thoracic diagnosis.
- COVID-19, TB and pneumonia labels can have different acquisition/annotation histories.
- A production prototype is not equivalent to clinical validation or regulatory approval.

23. Major Challenges and Mitigation

Challenge	Why it occurs	Mitigation
Dataset imbalance	Unequal class counts	Macro-F1, class weights, balanced sampling, targeted augmentation
Data leakage	Same patient/duplicate across splits	Patient-level splitting, duplicate checks
Domain shift	Different institutions/devices/populations	External validation, source-aware analysis, domain-robust methods
Overfitting	Limited or homogeneous data	Augmentation, regularization, early stopping, external test
High compute cost	Multiple CNNs and attention variants	Transfer learning, staged experiments, mixed precision, cloud/GPU when available
Black-box behavior	Distributed learned representations	Grad-CAM and systematic explanation analysis
Misleading accuracy	Class imbalance	Macro-F1, per-class recall, confusion matrix
Reproducibility	Experiments proliferate quickly	MLflow, configs, seeds, versioned artifacts
Model drift	Future data distribution changes	Monitoring, versioned models, re-evaluation
Clinical overclaiming	Prototype mistaken for diagnosis	Decision-support framing and explicit limitations

24. Production Architecture

The final validated model is served through a backend inference API. A web client submits a CXR image, the API validates the input, applies the exact preprocessing used during training, loads the registered production model, returns class probabilities, and optionally generates a Grad-CAM visualization.



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- Frontend: image upload, prediction display, confidence/probability view, Grad-CAM visualization.
- FastAPI: inference endpoint, validation, preprocessing orchestration and response schema.
- Model service: loads the registered model and performs inference.
- MLflow: tracks training runs, parameters, metrics, artifacts and model versions.
- Docker: reproducible packaging of the inference environment.
- Cloud: hosts the service and, where appropriate, model/artifact storage.
- CI/CD: runs tests and builds/deploys the service.

25. Proposed API

Endpoint	Purpose
GET /health	Service health check
POST /predict	Upload CXR and return class probabilities
POST /explain	Return prediction plus Grad-CAM visualization
GET /model-info	Return deployed model/version metadata

The API should validate file type, file size and image readability and should return structured error responses rather than exposing stack traces.

26. MLflow / MLOps Design

Every experiment should record enough information to reproduce the result.

- Dataset version or data manifest identifier
- Model architecture
- Attention configuration
- Image resolution
- Augmentation configuration
- Optimizer and learning rate
- Batch size and epochs
- Random seed
- Validation metrics
- Test metrics
- Confusion matrix and plots
- Model artifact/checkpoint



- Git commit/version

A model registry can then separate candidate models from the validated production model.

27. Software Engineering Practices

- Configuration-driven experiments rather than hard-coded notebook cells.
- Modular separation of data, preprocessing, model, training, evaluation, explainability and API code.
- Unit tests for preprocessing and API behavior.
- Integration tests for model loading and inference.
- Input validation and safe error handling.
- Logging of inference requests without storing sensitive patient information unnecessarily.
- Environment pinning and reproducible Docker builds.
- Git-based version control.
- CI/CD for tests and deployment.

28. Security, Privacy and Responsible AI

- Do not treat uploaded medical images as ordinary public files.
- Avoid retaining identifiable patient information unless explicitly required and appropriately governed.
- Use HTTPS in deployed environments.
- Restrict access to model-management endpoints.
- Separate development/test data from production user uploads.
- Document model limitations and intended use.
- Do not represent model confidence as clinical certainty.
- Do not market the prototype as a replacement for radiologists.

29. Complete Research Workflow

Literature → research gap → dataset selection → data audit → patient-level split → preprocessing → CNN baselines → attention variants → ablation → internal evaluation → Grad-CAM → external evaluation → identify weakness → propose modification → validate contribution → select final model → MLflow registration → Docker/FastAPI → cloud deployment → monitoring/re-evaluation.



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30. Expected Results

No numerical result should be promised before experiments are conducted. The expected output is a comparative evidence table showing whether each architecture and attention configuration improves relevant metrics.

Model	Macro-F1	TB Recall	COVID Recall	External F1	Latency	Parameters
ResNet baseline	To be measured	To be measured	To be measured	To be measured	To be measured	To be measured
DenseNet baseline	To be measured	To be measured	To be measured	To be measured	To be measured	To be measured
EfficientNet baseline	To be measured	To be measured	To be measured	To be measured	To be measured	To be measured
Best attention variant	To be measured	To be measured	To be measured	To be measured	To be measured	To be measured
Final proposed model	To be measured	To be measured	To be measured	To be measured	To be measured	To be measured

31. Current Literature-Based Gaps

The literature supports several broad gaps relevant to this project:

- Generalization across datasets and institutions remains a major concern.
- Public CXR datasets have heterogeneous acquisition and labeling characteristics.
- High internal performance can overstate real-world performance.
- Explainability is often demonstrated qualitatively and can be difficult to validate objectively.
- Architecture comparisons are frequently confounded by different datasets, preprocessing and evaluation protocols.
- Class imbalance and minority-class behavior require more than aggregate accuracy reporting.
- Production reproducibility and model lifecycle management are often treated separately from the research model.

A 2026 review also emphasizes that pretraining strategy and dataset provenance can influence reported performance substantially and calls for stronger transparency, evaluation methodology and systematic external validation. 穀cite學turn0search10傢

32. Key Existing Work You Must Know

Work	Why it matters to PneumoNet
Deep learning for chest X-ray analysis: A survey (2021)	Maps CXR DL tasks, datasets, caveats and research directions. 穀cite學turn0search0傢
Chest X-ray analysis empowered with deep learning: systematic review (2022)	Summarizes CXR classification methods, datasets, challenges and future directions. 穀cite學turn0search5學turn0search9傢



Explanatory classification of CXR images into COVID-19, Pneumonia and TB (2022)	Critical baseline: four-category CXR work with XAI/Grad-CAM already exists. 穀cite學turn0search2學turn0search4像
External Validation of Deep Learning Algorithms for Radiologic Diagnosis	Supports the importance of external validation and documents frequent external-performance degradation. 穀cite學turn0search7像

33. What Would NOT Count as Novelty

- Using EfficientNet by itself.
- Using DenseNet by itself.
- Adding Grad-CAM by itself.
- Adding a standard SE/CBAM block by itself.
- Comparing three CNNs without a meaningful research question.
- Deploying the model with FastAPI.
- Putting the model into Docker.
- Using MLflow.
- Building a web UI.

These are valuable technical components, but the research paper needs a defensible scientific contribution.

34. What Can Count as a Stronger Contribution

- A rigorously demonstrated attention configuration that improves a meaningful metric under matched conditions and remains robust externally.
- A new or adapted training strategy shown to improve cross-dataset robustness.
- A method that reduces clinically implausible attribution patterns and is evaluated quantitatively.
- A class-aware learning strategy that materially improves difficult/minority disease recall without unacceptable trade-offs.
- A systematic backbone-attention interaction study that reveals a reproducible relationship overlooked by prior comparisons.
- A reproducible benchmark/protocol that exposes large differences between internal and external performance.

35. Limitations

- Public datasets may not represent the intended clinical population.
- Labels may be weak, noisy or derived from reports rather than expert image review.



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- Different datasets may use different disease definitions.
- The four classes are mutually exclusive in the project formulation, while real patients can have multiple simultaneous findings.
- Grad-CAM is coarse and does not provide precise lesion segmentation.
- External validation may be limited by availability and label compatibility.
- Computational constraints may restrict the number of architectures/hyperparameters evaluated.
- Clinical usefulness cannot be established without prospective clinical evaluation.
- Deployment does not imply regulatory clearance.

36. Future Scope

15. Expand from four classes to broader multi-label thoracic findings.
16. Investigate lung-region segmentation before classification.
17. Study transformer or hybrid CNN-transformer architectures.
18. Explore self-supervised or domain-specific pretraining.
19. Investigate domain adaptation and domain generalization.
20. Evaluate uncertainty estimation and probability calibration.
21. Perform prospective or multi-institution clinical validation.
22. Add model drift and data-quality monitoring.
23. Investigate federated learning where institutional data cannot be centralized.
24. Build a more rigorous quantitative benchmark for explanation quality.
25. Study fairness and subgroup performance across available demographic metadata.
26. Evaluate edge/low-resource deployment if the final model must operate on constrained hardware.

37. Conclusion

PneumoNet should be positioned as a research framework rather than a single neural-network architecture. The central scientific workflow is to establish strong CNN baselines, introduce attention under controlled conditions, perform ablation, evaluate class-wise performance, inspect explanations using Grad-CAM, and test external generalization. The project then uses the evidence from these experiments to identify and validate a targeted research contribution.

The production component complements the research by making the final model reproducible and deployable: MLflow tracks experiments and artifacts, the model registry manages validated versions, FastAPI provides inference, Docker packages the service, and CI/CD plus cloud infrastructure support deployment.



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The strongest academic outcome is therefore not a preselected claim that 'attention is better.' It is a defensible chain of evidence showing what works, where it works, where it fails, why it may fail, and what modification the project contributes to address that weakness.

38. One-Page Viva Explanation

If asked 'What exactly are you doing?', answer:

"We are building a four-class chest X-ray classification system for Normal, Pneumonia, TB and COVID-19. First, we establish CNN baselines using architectures such as ResNet, DenseNet and EfficientNet. We then introduce channel and spatial attention mechanisms and compare them under the same data and evaluation protocol. We perform ablation studies to determine which attention component contributes to any improvement. We use Grad-CAM to analyze what regions influence the predictions and, where possible, evaluate the final models on an external dataset to study domain generalization. The research contribution is not assumed in advance; it is derived from the weakness identified through these experiments and then validated against the baseline. Finally, we productionize the validated model using MLflow, model versioning, FastAPI, Docker, CI/CD and cloud deployment."

39. Final Deliverables

- Literature review and research-gap matrix
- Cleaned/versioned dataset manifest
- Reproducible preprocessing pipeline
- CNN baseline implementations
- Attention implementations
- Experimental/ablation results
- Grad-CAM analysis
- External validation report where feasible
- Research contribution and validation
- MLflow experiment history
- Registered final model
- FastAPI inference service
- Docker image
- Automated tests and CI/CD pipeline
- Cloud-hosted prototype
- Technical documentation



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- Final research paper and presentation

40. Recommended Project Chapter Structure

27. Chapter 1 — Introduction and Motivation
28. Chapter 2 — Background and Literature Review
29. Chapter 3 — Problem Statement, Objectives and Research Questions
30. Chapter 4 — Dataset and Data Engineering
31. Chapter 5 — Proposed Methodology
32. Chapter 6 — CNN Baselines and Attention Mechanisms
33. Chapter 7 — Experimental Design and Evaluation
34. Chapter 8 — Grad-CAM and Explainability Analysis
35. Chapter 9 — External Validation and Generalization
36. Chapter 10 — Proposed Research Contribution
37. Chapter 11 — MLOps and Production Architecture
38. Chapter 12 — Results and Discussion
39. Chapter 13 — Limitations
40. Chapter 14 — Conclusion and Future Scope

41. Final Stress-Test of the Project

The strongest criticism of the project is that the basic combination—four-class CXR classification + transfer learning + attention + Grad-CAM—has already been explored. A 2022 paper already demonstrated four-category CXR classification with COVID-19, Pneumonia, TB and Normal and used multiple XAI techniques including Grad-CAM. [cited in search2 search4](#)

Therefore, the project survives academically only if it moves beyond that recipe. The strongest route is controlled experimentation plus external generalization and a contribution derived from a demonstrated weakness. If the final work only reports that EfficientNet + CBAM achieved a slightly higher accuracy than DenseNet, it should be presented as an engineering/experimental project rather than a strong novel research paper.

That is the standard this artifact is designed around.

42. Source and Literature Basis

The artifact is grounded in the project material supplied in the conversation and supplemented with current literature checks. Key literature includes the CXR deep-learning survey, a systematic review of CXR deep learning, the published four-category COVID/Pneumonia/TB/Normal explanatory classification study,



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and evidence on external validation in radiologic deep learning.

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